RSNet Pilot: Findings, the B-cell Immune Module, and Where to Go Next

METABRIC recurrence study | Prepared for Tri Nham to read and think | October 14, 2026

The short version

We tested whether gene modules learned by Dr. Monti's RSNet framework predict breast cancer recurrence better than the curated gene sets we already use (MSigDB Hallmark and Desmedt). **They did not.** RSNet modules added no predictive value in any patient group, and they performed the same as random groups of genes. The most interesting finding is biological rather than predictive: RSNet, without ever seeing patient outcomes, discovered a stable **B-cell-rich immune module**, and in ER-negative patients a higher score for that module goes with clearly lower recurrence. That module overlaps heavily with immune scores we already have, which is why it did not improve the model.

1. What we actually did

RSNet is a **resampling-based network method**. It resamples the patients 100 times, builds a gene-to-gene network (which genes move together after accounting for all others) each time, and keeps only the connections that keep showing up. A community-detection step (Leiden) then groups connected genes into **modules**. RSNet never sees recurrence outcomes; it only looks at gene expression.

Each module was turned into **one score per patient** (the first principal component of its genes, fitted only on training patients). We then ran the same models as the main study (tuned ridge Cox and random survival forest) on the same 1,295 METABRIC patients and the same 50 cross-validation splits, comparing five feature sets: clinical only; clinical + gene-set scores (the comparator); clinical + RSNet modules; clinical + both; and clinical + random modules of the same sizes (a control showing whether network structure matters at all). The resampling was used only to learn a reliable gene network. It did **not** create new patients or fix class imbalance.

To avoid information leaking between training and testing, modules used in METABRIC were learned on TCGA expression (direction A), and modules used for the TCGA external check were learned on METABRIC (direction B). All settings were written down before any result was seen (PREREGISTRATION.md); every change is logged (DEVIATIONS.md).

2. Main result

Population	Clinical only	+ Gene sets	+ RSNet modules	+ Both	+ Random modules
All (1,295 / 545 events)	0.653	0.660	0.654	0.659	0.654
ER-negative (299 / 137)	0.666	0.655	0.649	0.648	0.646
ER-positive (996 / 408)	0.632	0.660	0.636	0.657	0.634

Values are mean C-index from tuned ridge Cox over 50 splits (0.5 = chance, 1.0 = perfect ranking).

Pre-registered comparison	Population	Mean change	Splits won	Holm p
RSNet vs gene sets	All	-0.006	15 / 50	1.00
RSNet vs gene sets	ER-negative	-0.006	20 / 50	1.00
Both vs gene sets	All	-0.001	18 / 50	1.00
Both vs gene sets	ER-negative	-0.007	13 / 50	1.00
RSNet vs random modules	All	-0.001	26 / 50	1.00
RSNet vs random modules	ER-negative	+0.003	33 / 50	1.00

The external check on TCGA agreed: METABRIC-learned modules were worse than gene-set scores in all patients (C-index change -0.036, 95% CI -0.068 to -0.007) and indistinguishable in ER-negative patients (34 events, wide CI). Secondary analyses (random survival forest, 5-year time-dependent AUC, mean-expression scoring) all showed the same null pattern.

3. The B-cell immune module

What the output shows

The most stable module in the Cox models was **M7** (44 genes). It ranked among the five most influential modules in 94% of splits in all patients and 98% in ER-negative patients, and its coefficient was negative, meaning a higher score goes with lower recurrence risk. Its most connected (hub) genes are MS4A1, IL6, UBD, AQP9 and FCER2, and its strongest Hallmark overlap is **interferon-gamma response** (6 genes, significant after correction). Source: results/A1000_eig/biology_top_modules.csv.

Why we call it a B-cell module

This label is an interpretation of the gene list, checked against reference marker lists:

- **MS4A1** is the gene for CD20, the classic B-cell surface marker (the target of the drug rituximab). **FCER2** is CD23, also found on B cells.
- The module contains **5 of the 9 B-cell markers** in the MCP-counter reference list: CD19, CD79A, CR2, MS4A1 and PAX5.
- It also contains other B-cell and antibody-producing (plasma) cell genes: BLK, FCRL3, FCRL5, FCRLA, POU2AF1, SPIB, TCL1A, TNFRSF17 and CD38.
- Alongside these are interferon and inflammation genes (CXCL9, CXCL11, IDO1, IL6, GZMB, S100A8/9). The precise description is therefore **an immune module dominated by B cells, with interferon-gamma and inflammatory genes**.

What a B cell is

B cells are white blood cells of the immune system whose main job is to make antibodies that tag invaders for destruction. Everyone has them, mainly in blood, lymph nodes and the spleen. In cancer, immune cells, including B cells, can move into the tumor. When a tumor sample contains many of them, the genes above are highly expressed, so a high M7 score means a tumor with a strong immune presence. These immune cells are mostly there to attack the tumor; they do not shield it from treatment.

A closer look at M7 (exploratory, added for this report)

Population	Recurrence: low M7 third	middle third	high M7 third	Hazard ratio per SD	Adjusted HR per SD*
All	35.6%	49.0%	41.7%	1.00 (0.92-1.09), p = 0.98	0.91 (0.84-0.99), p = 0.035
ER-negative	56.0%	46.5%	35.0%	0.82 (0.70-0.97), p = 0.018	0.81 (0.69-0.94), p = 0.008
ER-positive	35.5%	42.5%	44.9%	1.05 (0.93-1.18), p = 0.41	0.97 (0.87-1.09), p = 0.65

*Adjusted for age, NPI and positive lymph nodes. Fitted on all patients; descriptive only, and M7 was singled out after seeing the results, so these p-values are not confirmatory. Source: results/A1000_eig/M7_descriptive.csv.

- **The protective association is an ER-negative story.** In ER-negative patients, recurrence falls from 56% in the lowest third of M7 to 35% in the highest third, and each standard deviation of M7 lowers the hazard by about 18-19%, even after adjusting for clinical factors.
- **In all patients combined the raw association is flat** because immune-rich tumors are more often ER-negative (M7 is highest in claudin-low and basal tumors and lowest in luminal A/B), and those recur more. After adjusting for clinical factors, the protective direction appears (HR 0.91).
- **In ER-positive patients there is no association.**
- **Why it did not improve prediction:** M7 correlates strongly with immune scores already in our dataset: r = 0.86 with the Desmedt immune (STAT1) module, 0.86 with MCP-counter B cells, 0.90 with MCP-counter T cells, and 0.79 with Hallmark interferon-gamma. The gene-set block already carried this signal.

How it fits the literature

This matches published biology: Desmedt et al. (2008) found the immune-response module was the main prognostic signal in ER-negative/HER2-negative tumors, and pooled analyses show tumor-infiltrating lymphocytes are strongly prognostic in triple-negative breast cancer (Loi et al., 2019). Ali et al. (2016) add an important nuance: the *type* of immune cell matters, with regulatory T cells and M2 macrophages linked to worse outcomes and CD8 T cells to better ones. The point worth emphasizing to Dr. Monti is that RSNet **recovered this known ER-negative immune biology from expression alone, without being given any outcome**, which supports the framework's ability to find coherent biology even though it did not beat curated gene sets for prediction.

4. Other things that stood out

- **In ER-negative patients the clinical model alone was best.** Adding any molecular block lowered the C-index (0.666 clinical, 0.655 with gene sets, 0.649 with RSNet, 0.646 with random modules). With 137 events, extra features mostly add noise. This is new information for the main study, which had shown genomic gains only in ER-positive patients.
- **Dr. Monti's hypothesis that networks might help outside ER-positive disease was not supported** in this setting.
- **Gene selection may be the bottleneck, not the network method.** In ER-positive patients gene sets added +0.028, while RSNet and random modules added almost nothing. Picking the 1,000 most variable genes likely misses the proliferation and ER-signaling genes that drive the ER-positive gain.
- **The two learned networks differ a lot.** The METABRIC network has about twice the edges of the TCGA network (2,085 vs 1,055), the two 1,000-gene lists share only 350 genes, and several TCGA modules are built from genes that are either fully on or off in RNA sequencing (for example cystatins and

spermatogenesis genes). Learning on one platform and testing on another probably hurt transfer.

5. Where we can go from here

Option A: report the null result now (recommended first step)

Send Dr. Monti the filled results template and a short summary: no predictive gain over gene sets or random modules; ER-negative patients are best served by the clinical model in this cohort; but RSNet recovered a stable, interpretable B-cell/interferon module whose protective association is concentrated in ER-negative disease. Mention the caveats (cross-platform learning, top-variance gene selection, 500-gene sensitivity still pending) and ask whether he would like a follow-up.

Option B: follow-ups that would test RSNet more fairly (new analyses, to be pre-registered)

- **Same-platform learning:** learn modules within METABRIC inside each training fold (or on a held-out half), so modules and evaluation share the same microarray platform.
- **Better gene pool:** restrict to well-expressed genes before choosing the most variable ones, or start from a biologically informed pool (for example the union of Hallmark genes), so proliferation and ER genes are included.
- **ER-negative immune deep dive:** split the immune signal by cell type (B cells, CD8 T cells, regulatory T cells, macrophages) and test which carries the ER-negative protective effect, following Ali et al. (2016).
- **Test M7 as a pre-specified biomarker in TCGA:** fix the M7 gene list and scoring now, then test its association with recurrence in TCGA ER-negative patients as an independent check.
- **Finish the 500-gene sensitivity runs** (two 10-minute RSNet runs on the Mac).

Option C: what this means for the main AMIA paper

- Add the ER-negative result: genomic and network features help ER-positive recurrence prediction but not ER-negative, where the clinical model is already as good as it gets with this sample size.
- Optionally include the RSNet comparison as a short section: a data-driven network method did not outperform curated gene sets, which strengthens the case that domain-knowledge features are the right choice for this task.
- Keep the claims modest: these are internal cross-validation results in one historical cohort, with a limited TCGA check.

6. Questions to think about

- Is the ER-negative immune signal better captured by B cells specifically, or by immune activity in general?
- Would RSNet look better if it learned networks inside METABRIC, on the same platform used for evaluation?
- Should the next step be a stronger prediction test, or a biology-focused paper on the ER-negative immune module?
- How much of this belongs in the AMIA paper versus a separate collaboration with Dr. Monti?

Files: pilot_rsnet/RESULTS.md (full numbers), PREREGISTRATION.md, DEVIATIONS.md, results/A1000_eig/, results/A1000_meanz/, results/B1000_external/, rsnet_output/.

References: Desmedt et al., Clin Cancer Res 2008; Loi et al., J Clin Oncol 2019; Ali et al., PLoS Med 2016; Becht et al. (MCP-counter), Genome Biol 2016; Liberzon et al. (MSigDB Hallmark), Cell Syst 2015; Huang, Song, Sebastiani, Monti, arXiv:2605.12706 (RSNet).